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#include <iostream>
using namespace std;

class complex {    // Defines a new class named 'complex'
    int real, imag; // Private member variables to store the real and imaginary parts
public:            // Access specifier: everything below this can be accessed from outside the class
    // Constructor
    complex(int r, int i) {
        real = r;
        imag = i;    // Assigns the input value 'i' to the member variable 'imag'
    }                // Ends the constructor definition
    // Member function to print the complex number to the screen
    void display() {
        if (imag >= 0) { // Checks if the imaginary part is positive or zero
            // Prints the format: "real + imaginary i" (e.g., 2 + i3)
            cout << real << " + i" << imag << endl;
        }
        else {    // Executes if the imaginary part is negative
            // Creates a temporary variable to turn the negative number into a positive one
            int positive_imag = -imag;
            // Prints the format: "real - imaginary i" (e.g., 4 - i6) instead of 4 + i-6
            cout << real << " - i" << positive_imag << endl;
        }
    }
};

int main() {
    complex c1(2, 3);
    cout << "The complex number is: ";
    // Calls the display method on object 'c1' to print its values
    c1.display();
    return 0;
}

```

Output

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The complex number is: 2 + i3
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